International Journal of Obstetric Anesthesia 66 (2026) 104849

Contents lists available at ScienceDirect

International Journal of Obstetric Anesthesia

journal homepage: www.elsevier.com/locate/ijoa

Brief Report

Postdural puncture headache after neuraxial labor analgesia with 25- vs. 27-gauge Whitacre needles: a single-center retrospective cohort study (2018–2023)

M. Balot *, G. Echevarria, A. Lee, B. Mahoney

Icahn School of Medicine, Department of Anesthesiology, Perioperative and Pain Medicine, Mount Sinai West and Morningside Hospitals, New York, NY, USA

ARTICLE INFO

Keywords:
Postdural puncture headache
Combined spinal epidural
Dural puncture epidural
Needle size
Spinal headache
Neuraxial Labor analgesia

ABSTRACT

Background: Combined spinal epidural (CSE) and dural puncture epidural (DPE) procedures have become increasingly utilized for labor analgesia, though no standards exist on which spinal needle to use. It has been demonstrated that pencil-point needles reduce the rate of postdural puncture headaches; however, there is no consensus on the optimal size to minimize headache risk while maximizing analgesia.

Methods: A single-center retrospective cohort study was conducted of parturients receiving CSE or DPE for neuraxial labor analgesia between 2018 and 2023 to identify the postdural puncture headache rate with intended (but not unintended) dural puncture.

Results: A total of 10,459 CSE and DPE procedures were performed during the study period, with CSE most commonly performed (91%). For CSE, 27G needle was used in 69% of cases, and for DPE, 25G needle was used in nearly two-thirds of cases. Fifty-four patients (0.52%) experienced a postdural puncture headache (without evidence of unintended dural puncture with epidural needle). Postdural puncture headache occurred more frequently with 25G needles than with 27G needles (0.73% vs. 0.41%), corresponding to a risk ratio of 0.56 (95% CI 0.34–0.92; P=0.029)).

Conclusions: There was a small but statistically significant increase in the incidence of postdural puncture headache when comparing rates with use of 25G vs. 27G needles during CSE or DPE. However, the clinical significance of this modest increase must be weighed against the potential benefit of improved analgesia with the use of larger-gauge (25G) pencil-point spinal needles.

Introduction

Neuraxial labor analgesia techniques with intentional dural puncture with an atraumatic pencil-point spinal needle, such as dural puncture epidural (DPE) or combined spinal epidural (CSE), are increasingly utilized.[1] However, there is concern that the intended dural puncture may increase the risk of postdural puncture headache (PDPH),[2] though large studies have reported a low incidence of PDPH after CSE or DPE.[2–4]

There is some limited data regarding differences in PDPH rates when 25G and 27G pencil-point needles are used for neuraxial labor analgesia.[5] A recent consensus statement suggested smaller non-cutting needles may reduce PDPH risk, but the evidence was of only moderate certainty.[6] This conflicts with a large meta-analysis that found no association between needle gauge for pencil-point needles and PDPH,[7] while a subsequent systematic review showed that 27G pencil-point needles do carry a lower risk of PDPH when compared to 25G needles, at the expense of reduced procedural success.[8] We sought to determine whether PDPH rates differed after use of atraumatic 25G and 27G needles for neuraxial labor analgesia or anesthesia. We hypothesized no difference in PDPH rates between needle sizes.

Methods

Following Institutional Review Board approval (21–01791), we reviewed the medical records of women who developed a PDPH after neuraxial labor analgesia between September 2018 to October 2023, excluding cases with unintented dural puncture (UDP) with the epidural needle.

We identified patients who received CSE or DPE, excluding standard epidurals and single-shot spinals. Procedure notes were queried for

* Corresponding author at: 1000 10th Avenue, New York, NY 10019, USA.
E-mail address: Michael.Balot@mountsinai.org (M. Balot).

https://doi.org/10.1016/j.ijoa.2026.104849
Accepted 9 January 2026
Available online 10 January 2026
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terms suggesting an UDP ("wet tap," "fluid through the Tuohy") and flagged records were excluded (Supplement 1).

To identify patients with PDPH without an UDP, follow-up and discharge notes were queried for proxy terms such as "headache," "PDPH," or "laying flat" (Supplement 1). All records with an epidural blood patch were reviewed. Use of acetaminophen/butalbital/caffeine (Fioricet®) was also considered a proxy for PDPH.

PDPH was defined as a positional headache presenting after delivery. Non positional headaches were excluded. Two authors (M.B. and G.E.) independently reviewed all records identified as PDPH without UDP. Inter-rater agreement for the initial independent review was high, agreeing on 49 of 54 cases (90.7%). For the five cases where assessments differed, the reviewers reached consensus after case discussion without the need for a third adjudicator.

We extracted mode of delivery, type of neuraxial technique, and spinal needle size. Braun PERIFIX® Continuous Epidural kits with 17G Tuohy needles and Braun PENCAN® 25G and 27G pencil-point spinal needles were used.

Statistical analysis

Categorical variables were compared using $\chi 2$ tests, and continuous variables were summarized as mean ± standard deviation (SD) or median with interquartile range [IQR], as appropriate.

A multivariable logistic regression model was used to evaluate the association between PDPH without UDP and needle size. Variables with $P < 0.25$ in univariable analyses were considered as candidate covariates for the multivariable model. Neuraxial technique (CSE vs. DPE) was prespecified as a potential confounder regardless of univariable significance.

Given the rarity of the outcome, adjusted risk ratios (aRRs) and adjusted risk differences (aRDs) with 95% confidence intervals (CIs) were estimated using marginal standardization of the fitted models. Stepwise backward elimination using likelihood ratio tests ($P < 0.20$) was applied to covariates identified through univariable screening, while the prespecified confounder (neuraxial technique) was retained. Model calibration was assessed with the Hosmer–Lemeshow goodness-of-fit test. All tests were two-sided, and a P value < 0.05 was considered statistically significant. Analyses were performed using R version 4.5.0 (https://www.R-project.org/).

Results

We identified 10,459 patients receiving CSE or DPE for labor analgesia; 91% of cases were CSE. Regarding needle size, 69% (6,547/9,491) of CSE procedures were performed using a 27G needle, whereas nearly two-thirds of DPE were performed with a 25G needle (Table 1). Baseline demographic data are shown in Table 2.

Fifty-four patients (0.52%) developed PDPH without unintended dural puncture with epidural needle. PDPH occurred more frequently with a 25G needle [0.73% (26/3,564)] than with a 27G needle [0.41% (28/6,895)] (Table 3). The absolute risk difference was 0.32% (95% CI 0.02–0.61%), and the risk ratio was 0.56 (95% CI 0.34–0.92), indicating a lower risk of PDPH with 27G needles ($P = 0.029$).

Univariable logistic regression results are shown in Table 4. In the primary multivariable analysis adjusting for gravidity, use of a 27G needle was associated with a lower risk of PDPH compared with a 25G needle (adjusted risk ratio 0.57, 95% CI 0.34–0.96; adjusted risk difference −0.31%, 95% CI −0.60 to −0.02%; $P = 0.041$). Neuraxial technique was evaluated as a prespecified confounder in sensitivity analyses. Inclusion of technique in the model resulted in a negligible change (1.1%) in the needle gauge effect estimate, indicating minimal confounding (Appendix A). Model calibration was adequate for both the primary and sensitivity models based on the Hosmer–Lemeshow goodness-of-fit test ($P = 0.08$ and $P = 0.076$, respectively).

During the study period, the incidence of UDP was 1.18%, for which an epidural blood patch was performed in 23.6%. In our cohort, of the 54 patients who developed PDPH without an UDP, 17 (31.5%) received a blood patch, with 34.6% in the 25G needle group and 28.6% in the 27G needle group.

Table 1
Distribution of needle gauge by neuraxial technique.

	25G needle	27G needle	Total
CSE	2,944 (31.0%)	6,547 (69.0%)	9,491 (100%)
DPE	620 (64.1%)	348 (35.9%)	968 (100%)
Total	3,564	6,895	10,459

Abbreviations: CSE, combined spinal epidural; DPE, dural puncture epidural.

Table 2
Patient characteristics and obstetric data.

	No PDPH (n=10,405)	PDPH (n=54)	P value
Age (years)	32.56 ± 5.11	32.04 ± 5.39	0.45
Pain score at request (0–10)	7.00 [4.00, 10.00]	7.00 [4.00, 9.50]	0.671
Cervical dilation at request (cm)	2.00 [1.00, 4.00]	3.00 [1.00, 5.00]	0.462
Gravidity	2.00 [1.00, 3.00]	1.00 [1.00, 2.00]	0.076
Delivery type			
Vaginal, n	8204 (78.8)	46 (85.2)	0.332
Intrapartum cesarean, n	2201 (21.2)	8 (14.8)	
Neuraxial technique			
CSE, n	9436 (90.7)	46 (85.2)	0.25
DPE, n	969 (9.3)	8 (14.8)	
Spinal needle gauge			
25G	3538 (34.0)	26 (48.1)	0.041
27G	6867 (66.0)	28 (51.9)	
Body mass index (kg/m^2)	29.78 ± 5.43	30.29 ± 5.52	0.496
Performed by (%)			
Resident, n	1768 (17.0)	8 (14.8)	0.808
Attending, n	8637 (83.0)	46 (85.2)	

Data presented as mean ± standard deviation (SD) or median with interquartile range [IQR], or number (%).

Table 3
Postdural puncture headache according to needle gauge.

	PDPH without UDP	No PDPH	Total
25G needle	26 (0.73%)	3,538 (99.27%)	3,564 (100%)
27G needle	28 (0.41%)	6,867 (99.59%)	6,895 (100%)
Total	54	10,405	10,459

Abbreviations: PDPH: postdural puncture headache; UDP: unintended dural puncture.

Table 4
Univariable analysis.

Variable	OR (95% CI)	P
Age (years)	0.98 (0.93–1.03)	0.45
Pain score at request	0.98 (0.90–1.07)	0.61
Cervical dilation at request	1.08 (0.94–1.24)	0.28
Gravidity	0.86 (0.68–1.09)	0.2
Delivery type	0.65 (0.31–1.38)	0.26
Neuraxial technique	0.59 (0.27–1.25)	0.17
Body mass index	1.02 (0.97–1.07)	0.5
Performed by resident	1.18 (0.55–2.50)	0.67

OR: odds ratio; CI: confidence interval.

Discussion

In this retrospective cohort study, 54 women had a PDPH after

neuraxial labor analgesia, of which 26 occurred with use of a 25G needle and 28 with a 27G needle, with an incidence of 0.73% and 0.41%, respectively. This represents the largest cohort to date examining the PDPH incidence during CSE or DPE placement with an atraumatic pencil-point spinal needle, providing further data to allow anesthesiologists to balance the risk of PDPH with superior labor analgesia.

Two studies directly compared 25G vs. 27G Whitacre spinal needles. In a cohort of 478 women receiving neuraxial labor analgesia with a CSE, the incidence of PDPH was 4% with a 25G needle compared with 0.7% with a 27G needle, which is congruent with the association found here but of a larger magnitude.[5] However, in a cohort of 800 patients, another study demonstrated no difference in PDPH incidence with the two needle sizes.[9]

Prior studies investigating the potential benefits of DPE technique have indicated that the advantages accompanying intentional dural puncture occur with larger gauge spinal needles (24-25G), while a decrease in unilateral blockade and superior sacral coverage is equivocal when utilizing 26 − 27G needles.[2,10] The superior analgesia is hypothesized to be due to greater subarachnoid flux of epidural-administered medication via the larger dural puncture hole.[2] Furthermore, larger needles may provide more stability in traversing ligaments and dural puncture, allowing for decreased technical difficulty and increased intentional dural puncture success rate.[8,11]

Our study revealed an association with increased epidural blood patch rates in the 25G group compared with the 27G group, both of which were higher than the blood patch rate in patients with UDP (34.6%, 28.6%, and 23.6%, respectively). This may reflect greater symptom burden in the 25G group; however, interpretation is limited due to the small number of PDPH cases, and the absence of standardized criteria for epidural blood patch, which is influenced by both anesthesiologist practice and patient preference. We also lack data on severity and duration of PDPH, a limitation of our retrospective study.

Our study has several other limitations. The retrospective design is subject to confounding by technique (CSE vs. DPE), provider preference, and stage of training. Attending anesthesiologists at our institution have a preference for spinal needle size based on their personal assessment of the literature; therefore, supervising anesthesiologists' bias may confound our findings. Additionally, there is likely ascertainment bias as documentation is limited regarding technical aspects of the neuraxial procedure, including number of attempts, levels attempted, and needle redirections. Search criteria used to identify PDPH may have been insufficient to capture all episodes that occurred, and assessment of potential PDPH occurred by a heterogeneous group of clinicians. Given the relatively small number of PDPH cases, and headaches resulting from spinal needle puncture would theoretically be milder, our study may substantially underreport the true incidence. Finally, the presumption that PDPH cases were due to dural puncture with a spinal needle (rather than an unrecognized UDP with the epidural needle) is almost certainly inaccurate and introduces an unknown degree of misclassification.

While further work is warranted to better characterize the impact of the size of pencil-point needles during CSE or DPE on PDPH incidence, our study provides the largest sample to date and demonstrates a small but statistically significant increased risk of PDPH with 25G spinal needles compared with 27G during CSE or DPE placed for labor analgesia. Anesthesiologists must weigh this potential small risk against the previously demonstrated analgesic benefits of using a larger pencil-point needle.

CRediT authorship contribution statement

M. Balot: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **G. Echevarria:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **A. Lee:** Writing – review & editing, Writing – original draft. **B. Mahoney:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijoa.2026.104849.

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